

# Geometry from Spline Interpolation

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## **Abstract**

To come.

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# Chapter 1

## Introduction

To come.

# Chapter 2

## Methods

### 2.1 Cubic Splines

Cubic splines are useful because of their inherent smoothness. The cubic spline, its first derivative, and its second derivative are all continuous. Starting with the cubic spline's third derivative, discontinuity arises.

Let a collection of  $n + 1$  points  $(x_i, y_i)$  for  $i = 0, 1, \dots, n$  approximate the function  $y = f(x)$  over the domain  $[x_0, x_n]$ . In total, there are  $n + 1$  points and  $n$  intervals. Let  $S(x)$  be the piecewise continuous spine, cubic in this case, that threads through each of the points  $(x_i, y_i)$ . The spline takes on  $n$  interval definitions as follows:

$$S_i(x) \triangleq a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3 \quad \text{for } x \in [x_i, x_{i+1}] \text{ and } i = 0, 1, \dots, n - 1. \quad (2.1)$$

The  $n$  foregoing definitions, with the coefficients  $a_i, b_i, c_i, d_i$ , give rise to  $4n$  (currently unknown) parameters used to define the complete spline  $S(x)$ .

Next we endow the spline with constraints on continuity and differentiability, which allow the  $4n$  unknown parameters to be solved. First, function continuity requires that the spline, on a particular interval, thread through collection of points  $(x_i, y_i)$  and the beginning and end of each interval. Thus,

$$S_i(x_i) = y_i \text{ and } S_i(x_{i+1}) = y_{i+1} \quad \text{for } i = 0, 1, \dots, n - 1. \quad (2.2)$$

This is the **continuity constraint**, which gives rise to  $2n$  conditions<sup>1</sup> and creates a spline that is piecewise continuous. Next, consider two **differentiability constraints**, which state that the first and second derivatives at a point  $x_i$  are continuous. That is, the first and second derivatives approaching a point  $x_i$  from the interval to the left side of the point must match the respective first and second derivatives approaching the point  $x_i$  from the interval to the right side of the point:

$$S'_{i-1}(x_i) = S'_i(x_i) \text{ and } S''_{i-1}(x_i) = S''_i(x_i) \quad \text{for } i = 1, 2, \dots, n - 1. \quad (2.3)$$

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<sup>1</sup>Two conditions per interval times  $n$  intervals yields  $2n$  conditions.

Because they apply only to the interior points (and not the endpoints  $x_0$  and  $x_n$ ), the differentiability constraints give rise to  $2(n - 1)$  constraints.<sup>2</sup>

Finally, two more conditions are necessary to completely define the spline  $S(x)$ . The two conditions are obtained by specifying differentiability conditions on the two spline endpoints  $x_0$  and  $x_n$ . Frequently, the first derivative is set to the known value of the function's derivatives at the endpoints,

$$S'_0(x_0) = f'(x_0) \text{ and } S'_{n-1}(x_n) = f'(x_n). \quad (2.4)$$

Alternatives include specifying the second derivative instead of the first,

$$S''_0(x_0) = 0 \text{ and } S''_{n-1}(x_n) = 0. \quad (2.5)$$

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<sup>2</sup>Two conditions per interior point times  $n - 1$  interior points yields  $2(n - 1)$  conditions.